

# RELTWIN: CONTRASTING LOCALLY VALID TIMESTAMP ANSWERS FOR QUERY-SPECIFIC AUDIO GROUNDING

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## ABSTRACT

Audio temporal grounding must distinguish which event occurrence answers a query, even when opposite event orders coexist in one recording. Models can instead return the same window for inverse queries, revealing a query-window binding error. We introduce RelTwin, an add-on fine-tuning method that constructs opposite-order windows from shared audio excerpts and contrasts their timestamp answers under each query. Both windows have acoustic support; the requested order determines the correct answer. This answer-level competition complements sequence supervision and preserves direct timestamp generation at inference. Applied to SpotSound-A, RelTwin reaches 59.43, 86.68, and 74.03 mIoU on SpotSound-Bench, Clotho-Moment, and UnAV-100, respectively. Against matched paired SFT, it improves joint localization by 31.25 points on synthetic relation-development data and SpotSound mIoU by 1.02 points. The binding advantage persists under timing intervention, supporting explicit supervision of which occurrence answers the query.

**Index Terms**— Audio temporal grounding, timestamp answer contrasts, query-window binding, paired supervision

## 1. INTRODUCTION

Audio temporal grounding maps a query to event intervals [1, 2]. In a recording containing both “dog then rooster” and “rooster then dog,” the descriptions refer to different windows. Recognizing the categories establishes presence; answering the query requires selecting the requested order. Returning the same real window for both queries is a *query-window binding error*. We target this distinction between event detection and query-specific localization.

RelTwin (Fig. 1) addresses this binding problem through *locally valid timestamp contrasts*. Shared excerpts form opposite-order windows in one recording. Each window contains a real event sequence, but its correctness as an answer depends on the query. Sequence supervision rewards correct timestamp answers individually; the added candidate loss

explicitly compares the two answers under each query. Reversing the query exchanges their labels while keeping the recording fixed, directly training the model to choose the requested occurrence.

RelTwin fine-tunes an existing grounding model using its own timestamp-answer likelihoods, preserving direct generation at inference. We evaluate the adapted SpotSound-A model on three public benchmarks and examine answer-level competition through matched training comparisons.

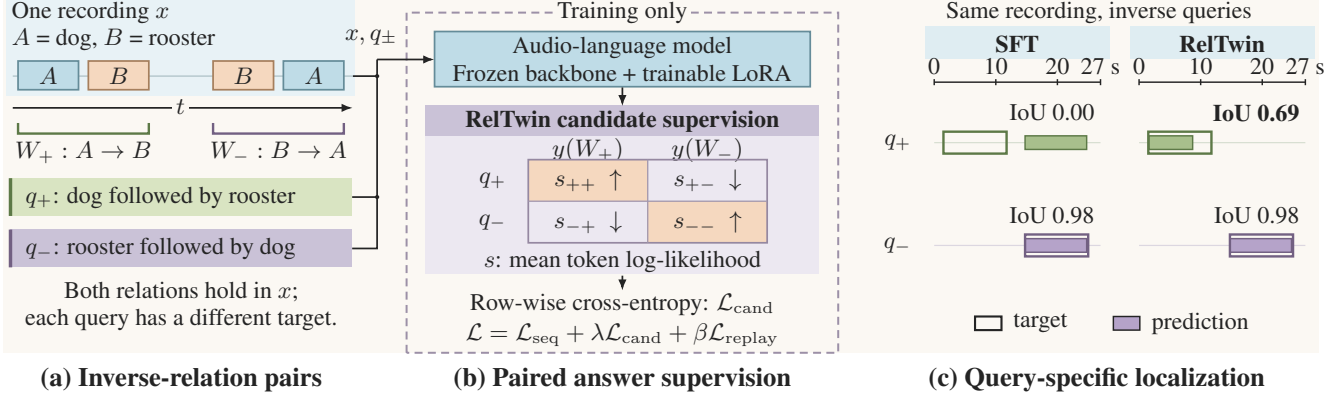
We make three contributions. First, we characterize query-window binding errors when inverse relations co-occur in one recording. Second, we introduce locally valid timestamp contrasts as an additional fine-tuning objective for query-specific answer selection. Third, public grounding evaluations, paired diagnosis, and a timing intervention connect the observed localization performance to improved query-window binding.

## 2. RELATED WORK

**Audio temporal grounding.** Weakly supervised grounding learns temporal correspondences from clip-level captions, reducing dependence on frame-level annotations [3]. Audio-language models such as Pengi [4], Qwen2-Audio [5], and Audio Flamingo 3 [6] provide audio-conditioned text generation. For temporal localization, TimeAudio introduces temporal markers and absolute time-aware encoding [7], while SpotSound interleaves timestamps with audio tokens and uses absent-event queries to suppress hallucinated intervals [8]. SpotSound thus combines event-presence supervision with timestamp prediction. RelTwin targets query-specific selection when inverse relations coexist: both descriptions have acoustic support in the recording, and the requested relation determines which occurrence is the answer.

**Compositional audio-language learning.** CLAP aligns audio and text in a shared representation space [9]. T-CLAP introduces order-reversed descriptions and a temporal-focused contrastive objective [10]. CompA evaluates event order and attribute binding, and trains CompA-CLAP with composition-aware hard negatives and modular contrastive learning [11]. CoSTALA combines spatio-temporal con-

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**Fig. 1.** RelTwin binds inverse queries to different windows in one recording. (a) Reused excerpts form two local targets. (b) The dashed enclosure marks training-only computation; shaded diagonal cells identify correct timestamp answers, and up/down arrows indicate training preferences. (c) In a relation-development example, SFT repeats one answer while RelTwin selects the corresponding window for each query. Outlines mark targets; filled bars show predictions, evaluated by set-IoU.

trast with local event-caption alignment and feature consistency [12]. These methods organize audio-text representations around compositional relations, including their local structure. RelTwin places the supervision on complete timestamp answer sequences: their query-conditioned likelihoods determine the preference between competing windows. This trains relational selection in the output space of a generative grounding model.

**Hard negatives for temporal grounding.** In compositional video grounding, SHINE combines generated hard-negative queries with within-video and cross-query saliency ranking [13]. Its training compares segment relevance as well as query compositions. RelTwin makes complete timestamp answers the candidates for comparison within one recording. Each candidate denotes an entire event sequence with real acoustic support. Under a fixed query, one answer matches the requested order and the other answers the inverse relation; reversing the query exchanges their labels while preserving the audio context. Comparing the length-normalized sequence log-likelihoods explicitly binds each relation query to its corresponding local answer.

### 3. QUERY-SPECIFIC WINDOW LEARNING

#### 3.1. Two true relations, two different answers

In one recording  $x$ , RelTwin arranges the same ESC-50 excerpts [14],  $A$  and  $B$ , as  $A \rightarrow B$  and  $B \rightarrow A$ . Their local targets  $W_+$  and  $W_-$  run from the first excerpt’s onset to the second’s offset, including the gap. Queries  $q_+$  and  $q_-$  request the respective orders. Both windows have acoustic support, but  $W_-$  is positive for  $q_-$  and negative for  $q_+$ ; reversing the query exchanges their labels.

The templates “followed by” and “and then” yield two inverse-query pairs per recording. Earlier-order layouts alternate (Section 5.3). Category rehearsal targets the union of

both occurrences of  $A$  or  $B$ , retaining category-grounding supervision.

#### 3.2. Contrast timestamp answers during training

We serialize an interval set  $S$  as comma-separated spans, each formatted like `from 1.500s to 11.750s` (three-decimal precision). Its supervised assistant-completion tokens  $y(S)$  exclude the prompt and are scored by

$$s_\theta(x, q, S) = \frac{1}{|y(S)|} \sum_t \log P_\theta(y_t | x, q, y_{<t}). \quad (1)$$

The  $2 \times 2$  matrix  $M$ , with entries  $M_{ij} = s_\theta(x, q_i, W_j)$  for  $i, j \in \{+, -\}$ , indexes queries by rows and answers by columns. Its diagonal labels  $Y = I_2$  identify the correct query-answer pairs (Fig. 1b). Sequence supervision minimizes  $\mathcal{L}_{\text{seq}} = -(M_{++} + M_{--})/2$ , rewarding the two correct responses individually. It does not explicitly compare them with the other locally valid window. RelTwin trains the relative preferences  $M_{++} > M_{+-}$  and  $M_{--} > M_{-+}$  through row-wise probabilities  $p_i(j) = \text{softmax}_j(M_{ij}/\tau)$ , where  $\tau > 0$ . The candidate loss is

$$\mathcal{L}_{\text{cand}} = -\frac{1}{2}[\log p_+(+) + \log p_-(-)]. \quad (2)$$

For a separately sampled category example  $(x_r, q_r, S_r)$ , the rehearsal loss is  $\mathcal{L}_{\text{replay}} = -s_\theta(x_r, q_r, S_r)$ . Together,

$$\mathcal{L} = \mathcal{L}_{\text{seq}} + \lambda \mathcal{L}_{\text{cand}} + \beta \mathcal{L}_{\text{replay}}. \quad (3)$$

Paired SFT sets  $\lambda = 0$ ; RelTwin uses  $\lambda > 0$  with identical relation and rehearsal data. Both directly generate timestamps at inference, without candidates or an additional scorer.

#### 3.3. Joint localization and relation selection

For interval unions,  $\text{IoU}_{\text{set}}(S, G) = |S \cap G|/|S \cup G|$  (zero for an empty union). Given predictions  $\widehat{W}_i$ , define  $u_{ij} =$

**Table 1.** Audio temporal grounding (%; higher is better). Baselines are quoted from SpotSound Table 3 [8]. RelTwin uses one adapted SpotSound-A checkpoint across the three benchmarks;  $\Delta$  subtracts the corresponding published base-model row, in percentage points. Bold marks the highest reported score per column; – denotes pending results.

| Method                                           | SpotSound-Bench |              |              | Clotho-Moment |              |              | UnAV-100 subset |              |              |
|--------------------------------------------------|-----------------|--------------|--------------|---------------|--------------|--------------|-----------------|--------------|--------------|
|                                                  | R1@.3           | R1@.5        | mIoU         | R1@.3         | R1@.5        | mIoU         | R1@.3           | R1@.5        | mIoU         |
| <i>Non-LLM models</i>                            |                 |              |              |               |              |              |                 |              |              |
| WTATG                                            | 53.0            | 34.5         | 38.4         | 12.1          | 6.3          | 9.1          | 53.0            | 37.0         | 38.4         |
| AM-DETR                                          | 25.5            | 15.0         | 19.5         | 89.8          | 88.0         | 80.9         | 59.0            | 46.0         | 42.8         |
| <i>Proprietary models</i>                        |                 |              |              |               |              |              |                 |              |              |
| Gemini-2.5-Flash                                 | 34.8            | 27.8         | 25.7         | 45.7          | 33.7         | 36.9         | 48.0            | 37.0         | 35.6         |
| Gemini-2.5-Pro                                   | 20.0            | 17.3         | 19.7         | 40.4          | 32.5         | 32.5         | 39.0            | 35.0         | 34.6         |
| <i>Open-source models and RelTwin adaptation</i> |                 |              |              |               |              |              |                 |              |              |
| Kimi-Audio                                       | 3.0             | 1.3          | 3.2          | 0.7           | 0.1          | 0.9          | 8.0             | 2.0          | 5.3          |
| Qwen2-Audio                                      | 5.5             | 1.8          | 4.4          | 6.1           | 1.3          | 5.7          | 14.0            | 4.0          | 9.7          |
| Audio Flamingo 3                                 | 8.8             | 2.8          | 7.6          | 32.9          | 21.8         | 22.6         | 35.0            | 24.0         | 25.0         |
| TimeAudio                                        | 7.8             | 1.8          | 10.2         | 39.5          | 24.9         | 28.6         | 21.0            | 7.0          | 16.0         |
| TimeAudio + RelTwin                              | –               | –            | –            | –             | –            | –            | –               | –            | –            |
| $\Delta$ vs. TimeAudio                           | –               | –            | –            | –             | –            | –            | –               | –            | –            |
| SpotSound-A                                      | 76.5            | 59.5         | 57.9         | 93.4          | 91.0         | 85.6         | 86.0            | 74.0         | 69.8         |
| SpotSound-A + RelTwin                            | <b>78.00</b>    | <b>62.00</b> | <b>59.43</b> | <b>93.61</b>  | <b>91.01</b> | <b>86.68</b> | <b>89.00</b>    | <b>79.00</b> | <b>74.03</b> |
| $\Delta$ vs. SpotSound-A                         | +1.50           | +2.50        | +1.53        | +0.21         | +0.01        | +1.08        | +3.00           | +5.00        | +4.23        |

$\text{IoU}_{\text{set}}(\widehat{W}_i, W_j)$ . PairAcc requires  $\min(u_{++}, u_{--}) \geq 0.5$ . Swap error occurs if  $u_{++} \leq u_{+-}$  or  $u_{--} \leq u_{-+}$ , including ties. JointPairAcc requires PairAcc success without Swap error. We report pair-level percentages; same-answer counts use exact equality of parsed interval lists.

Predicting the union of two disjoint, equal-length targets for both queries passes PairAcc (IoU 0.5) but fails JointPairAcc. Inspired by paired compositional evaluation [15], JointPairAcc distinguishes query-specific selection from coverage.

#### 4. EXPERIMENTAL SETUP

**Data and evaluation protocol.** Adaptation uses 40 ESC-50 classes and 256 constructed recordings: 1,024 relation queries (512 pairs) and 512 rehearsal queries. Ten held-out classes provide 80 recordings, 320 queries, and 160 inverse pairs for relation development. Training and development use disjoint classes and original Freesound IDs at the adaptation stage. Public evaluation uses all 400 SpotSound-Bench queries over 387 audios, 6,649 Clotho-Moment queries, and the released UnAV-100 subset with 100 queries over 77 audios [2]. We report set-mIoU and R1 at IoU thresholds 0.3 and 0.5, clipping intervals to audio duration. Relation-development and public-benchmark scores informed method development.

**Training settings.** SFT and RelTwin initialize from SpotSound-A on Audio Flamingo 3. LoRA [16] uses rank 8,  $\alpha = 16$ , dropout 0.1, and 20.19M parameters. Both run 256 AdamW [17] updates at learning rate  $5 \times 10^{-6}$ , zero weight decay, gradient clipping 1.0,  $\tau = 1$ ,  $\beta = 0.5$ , and RelTwin

$\lambda = 1$ . Each update accumulates one inverse-query pair and one rehearsal query. The matched comparison shares initialization, data order, code, environment, and optimizer updates. RelTwin additionally computes four gradient-free candidate scores followed by weighted gradient recomputation, versus SFT’s two positive-answer backpropagations. Both include rehearsal.

**Inference.** SFT and RelTwin are evaluated after 256 updates. All generators use 16 kHz mono audio, bfloat16, and greedy one-beam decoding without sampling, capped at 128 new tokens. The post-development timing diagnostic uses the frozen matched checkpoints (Section 5.3).

**Comparison roles.** Table 1 quotes published baselines;  $\Delta$  subtracts the corresponding base-model row. SpotSound-A + RelTwin uses one seed-0 checkpoint across all three benchmarks. Table 2 compares the official backbone, paired SFT, and RelTwin under shared inference. The backbone scores 58.42 SpotSound mIoU versus published 57.9; RelTwin’s three-seed gain against this re-evaluation is +0.75 points (audio-group CI  $[-0.24, 1.74]$ ).

**Result reporting and statistics.** RelTwin point results use seed 0, selected by SpotSound mIoU from five runs, with its matched SFT control. Table 2(c) separately summarizes seeds 0/1/2 as mean  $\pm$  sample SD. Training comparisons use 20,000 audio-group bootstrap resamples for 95% CIs. Relation comparisons resample 39 connected source-sharing groups (largest: 18 audios). CIs quantify audio-sample uncertainty conditional on the evaluated checkpoints; model selection is fixed across resamples.

**Table 2.** SpotSound transfer and synthetic relation-development binding. (b) Matched data and updates; (c) mean  $\pm$  sample SD (Section 4). Joint: JointPairAcc; same-answer: inverse-query pairs with identical timestamp answers. Scores are percentages except counts.

| Method                                                    | SpotSound        | Relation development (160 inverse-query pairs) |                  |                         |                                |
|-----------------------------------------------------------|------------------|------------------------------------------------|------------------|-------------------------|--------------------------------|
|                                                           | mIoU $\uparrow$  | mIoU $\uparrow$                                | Joint $\uparrow$ | Swap error $\downarrow$ | Same-answer pairs $\downarrow$ |
| <i>(a) Backbone reference: shared evaluation protocol</i> |                  |                                                |                  |                         |                                |
| SpotSound-A                                               | 58.42            | 37.64                                          | 0.63             | 65.00                   | 15/160                         |
| <i>(b) Matched training comparison</i>                    |                  |                                                |                  |                         |                                |
| Paired SFT                                                | 58.42            | 74.58                                          | 53.13            | 40.00                   | 37/160                         |
| RelTwin                                                   | <b>59.43</b>     | <b>88.55</b>                                   | <b>84.38</b>     | <b>7.50</b>             | <b>1/160</b>                   |
| RelTwin – Paired SFT                                      | +1.02            | +13.97                                         | +31.25           | −32.50                  | −36 pairs                      |
| <i>(c) RelTwin replication: three training seeds</i>      |                  |                                                |                  |                         |                                |
| RelTwin                                                   | 59.16 $\pm$ 0.26 | 87.05 $\pm$ 1.43                               | 81.46 $\pm$ 2.53 | 8.13 $\pm$ 0.63         | (1, 0, 0)/160                  |

## 5. RESULTS AND ANALYSIS

### 5.1. Grounding performance across benchmarks

SpotSound-A + RelTwin reaches 59.43, 86.68, and 74.03 mIoU on SpotSound-Bench, Clotho-Moment, and UnAV-100 (Table 1). The gains over the published base model are 1.53, 1.08, and 4.23 points. SpotSound recall increases by 1.50/2.50 points at IoU 0.3/0.5; UnAV recall increases by 3.00/5.00 points. These gains extend to the stricter localization criterion. Clotho recall reaches 93.61/91.01 versus published 93.4/91.0. One adapted checkpoint provides all three results.

On the same 400 SpotSound queries, the gain over matched paired SFT is 1.02 mIoU points, with audio-group CI [0.23, 1.81]: 92 wins, 239 ties, and 69 losses. The three-seed mean is 59.16 (Table 2c). The following analyses examine how candidate supervision changes the response to inverse queries.

### 5.2. Timestamp contrasts recover query-specific selection

Paired SFT returns identical answers for 37/160 inverse-query pairs across 22 recordings; RelTwin reduces this to one (Table 2). Of those 37 pairs, 27 become jointly correct: the two predictions recover their respective requested relations. Figure 1 illustrates this query-specific recovery, with each inverse query selecting its corresponding event window.

JointPairAcc increases by 31.25 points (source-group CI [22.35, 40.50]) and Swap error decreases by 32.50. Across all pairs, 51 become jointly correct and one regresses. Relation mIoU increases by 13.97 points (CI [9.84, 17.51]), connecting selection with overlap.

Across three RelTwin seeds, JointPairAcc averages 81.46 and identical-answer counts are 1, 0, and 0, reproducing the reduction in same-answer collapse.

### 5.3. Binding gains persist after timing intervention

To test layout-specific silence, we recompose 80 development audios with 1.5 s lead-in, 0.25 s within-window gaps, 3 s between windows, and 2 s tail, preserving waveforms, order, and queries. The frozen matched models evaluate one template (160 queries). Timing changes for 40 BA-first recordings; 40 AB-first recordings remain waveform-identical.

On changed recordings, RelTwin reaches 87.50 JointPairAcc against SFT’s 67.50 (+20.00; CI [2.94, 33.33]); originally they scored 85.00 and 42.50. RelTwin retains a 20-point selection advantage under the altered timing, supporting query–window binding across both silence layouts. Across all 160 queries, SFT and RelTwin score 81.34 and 90.65 mIoU.

### 5.4. Why locally valid answers matter

Shared excerpts hold event identity fixed while inverse queries exchange the correct window. Detecting both categories supports both candidates; selecting the requested order requires query-conditioned comparison. RelTwin’s reduced same-answer collapse and improved JointPairAcc capture this distinction in the controlled binding test. Timestamp-answer supervision preserves direct generation at inference.

## 6. CONCLUSION

RelTwin fine-tunes grounding models through locally valid timestamp contrasts: acoustically supported windows compete according to the query. The adapted SpotSound-A model exceeds its published mIoU on three benchmarks. Matched diagnosis and timing intervention demonstrate improved query–window binding. These results support explicit supervision of which occurrence answers the query while preserving direct timestamp generation at inference.

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**Compliance with Ethical Standards.** We reuse public ESC-50 [14], SpotSound [8], Clotho-Moment, and UnAV [2] data, including programmatically constructed sequences. No new recordings, participant recruitment, human or animal experiments, or listening studies were undertaken.

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